

Milestone 2 Report

Navigation Simulation: Comparison of Kalman Filter and Particle Filter for Robot Localization

CPE 593WS - Applied Data Structures and Algorithms

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Abstract

This project focuses on simulating robot localization in a 2-D dimensional environment using two estimation algorithms: Kalman and Particle Filters. The goal is to compare how both algorithms estimate

a robot's position when sensor measurements contain noise. In the current version of this project, a C++ simulation has been started where a robot moves with a constant velocity, and noisy measurements are generated, and both filters attempt to estimate the robot's position over time. The Kalman Filter is implemented using a simplified prediction and correction approach, while the Particle Filter uses multiple particles moving with noise which are weighted based on their distance from the sensor measurement.

Background

Robot localization is the process of estimating where a robot is located within a predefined space, or environment. This is relevant as robots are heavily dependent on sensors, however sensor data is rarely perfect. Measurements often contain noise due to environmental conditions, sensor limitations, or movement errors. Because of this, localization algorithms are implemented to estimate the most likely position of the robot.

The Kalman Filter is used most commonly in problems involving noisy data. It works best when the system is approximately linear and the noise obeys a Gaussian distribution. It uses two main steps: prediction and update. In the prediction step, the filter estimates the position of the robot based on its previous state and motion model. In the update step, it corrects that estimate using the latest sensor measurement.

The Particle Filter is another probabilistic localization method. Instead of keeping only one estimated state, it uses many possible states called particles. Each particle represents a possible robot position. Particles are moved according to the motion model, and are weighted based on how close they are to the measurements, and then used to estimate the position of the robot. Particle Filters are more useful when systems present nonlinear behavior, or deal with more complex uncertainties.

Recent robotics localization research continues to use Kalman-based and Particle-based methods for navigation and sensor fusion. For example, recent work has explored Kalman filtering for LiDAR localization and mobile robot localization, while Particle Filter approaches are still widely used in Monte Carlo localization and SLAM-related problems.

System Architecture

1. *Robot Motion Model*

The robot moves in the x and y directions with constant velocity.

2. *Sensor Measurement Model*

The simulated sensor observes the robot's position, but noise is added to represent real-world measurement error.

3. *Kalman Filter Estimator*

A simplified Kalman-style filter predicts the next position and corrects the prediction using the noisy measurement.

4. *Particle Filter Estimator*

A group of particles is created. Each particle moves with noise, receives a weight based on its distance from the measurement, and contributes to the final position estimate.

5. *Output System*

The program prints the true position, noisy measurement, Kalman estimate, and Particle Filter estimate for each time step.

Design and Algorithm

Kalman Filter Design

The Kalman Filter implementation uses a simplified version of the prediction and update process. First, the previous estimated position is updated using the robot's velocity. This gives a predicted position. Then, the filter compares the predicted position with the noisy measurement and corrects the estimate using a Kalman gain.

In this version, a fixed Kalman gain of 0.5 is used to keep the implementation simple. This means the filter balances the prediction and measurement equally. Although this is not a full matrix-based Kalman Filter yet, it demonstrates the main idea of the algorithm: combine prediction and measurement to reduce uncertainty.

Particle Filter Design

The Particle Filter implementation begins by creating 100 particles near the starting position. Each particle represents a possible estimate of the robot's location. At each time step, all particles are moved using the same velocity as the robot, but with additional random noise.

After movement, each particle is compared to the noisy sensor measurement. Particles closer to the measurement receive higher weights. The final particle estimate is calculated as a weighted average of all particles. This allows the filter to represent uncertainty using multiple possible positions instead of only one estimate.

Testing and Simulation Results

Testing is currently in progress. The simulation runs for 50 time steps. At each step, the robot moves diagonally in the 2D environment. The noisy measurement is usually close to the true position, but it fluctuates because of random sensor noise.

The early results show that the Kalman Filter produces a smoother estimate than the noisy measurements because it combines the predicted position with the measured position. The Particle Filter also follows the robot's path by using multiple weighted particles. Since this version is simplified, the results are mainly used to confirm that both algorithms are implemented and producing reasonable estimates.

More testing will be completed by comparing the average error of each filter. The error will be calculated as the distance between the true robot position and each filter's estimated position. This will make it possible to compare which method performs better under the same noise conditions.

Challenges

One challenge in this project is implementing the algorithms in a way that is simple enough for the project timeline but still represents the main concepts correctly. A full Kalman Filter usually uses matrices for state transition, measurement update, covariance, and uncertainty. For this milestone, the implementation uses a simplified Kalman gain approach. Another challenge is that a complete Particle Filter normally includes resampling, but the current version mainly uses weighted averaging. Resampling may be added in the next version.

Conclusion

At this stage, the project has moved from research and planning into implementation. A C++ simulation has been created to model robot movement, noisy sensor readings, and position estimation using both Kalman Filter and Particle Filter methods. The Kalman Filter provides a smoother estimate by correcting predicted motion with measurements, while the Particle Filter estimates position using multiple weighted particles.

The project is not fully complete yet, but the main functions are implemented and the simulation framework is working. The next steps are to improve the accuracy of both filters, add error calculations, create graphs from the output, and compare the performance of both algorithms more clearly.

References

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